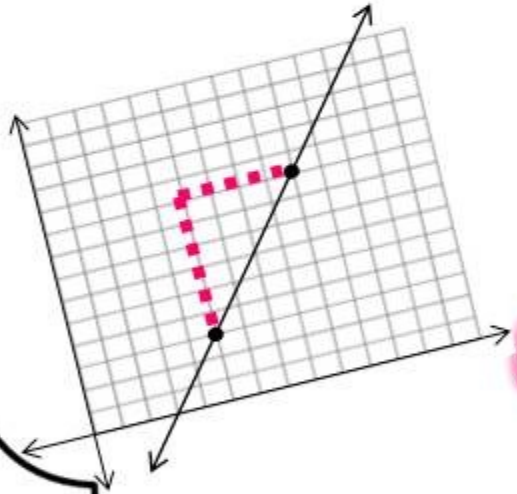
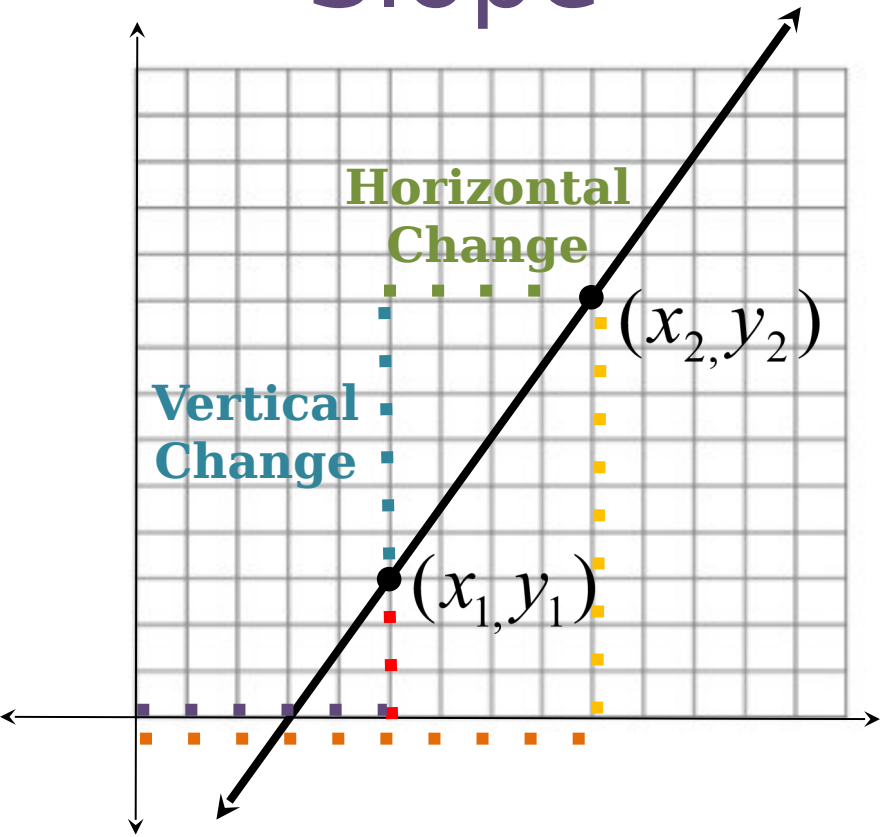
Slope



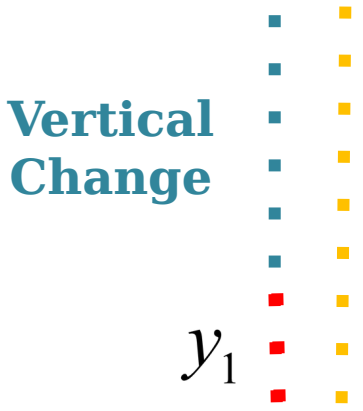
Using the slope formula

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Slope

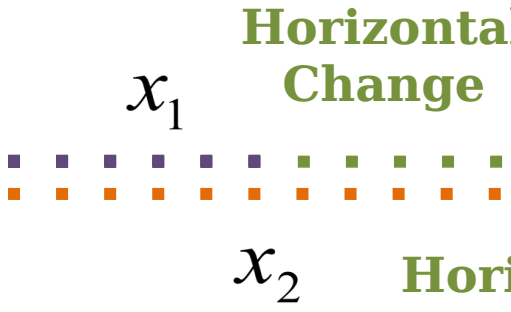


To Find Vertical Change



$$\text{Vertical Change} = y_2 - y_1$$

To Find Horizontal Change



$$\text{Horizontal Change} = x_2 - x_1$$

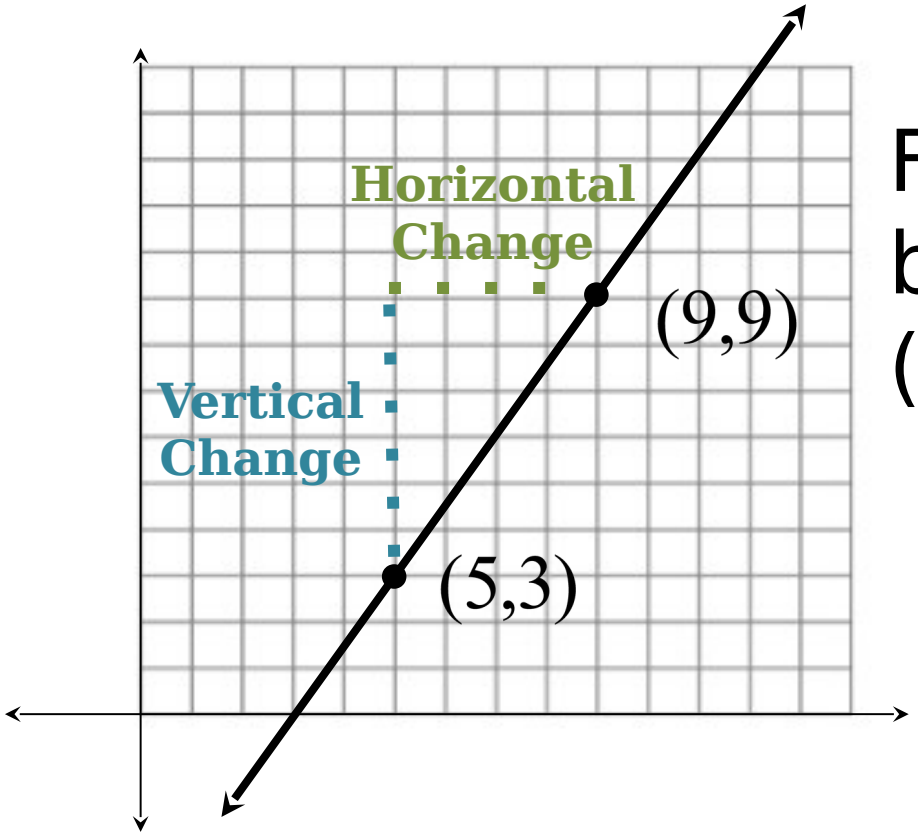
What are the lengths of the yellow and red lines?

What are the lengths of the orange and purple lines?

$$\text{Slope} = \frac{\text{Vertical Change}}{\text{Horizontal Change}} = \frac{(y_2 - y_1)}{(x_2 - x_1)}$$

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Example



Find the slope between (5, 3) and (9,9)

$$\text{Slope} = \frac{\textbf{Vertical Change}}{\textbf{Horizontal Change}} = \frac{(y_2 - y_1)}{(x_2 - x_1)}$$

$$= \frac{(9-3)}{(9-5)} = \frac{6}{4} = \frac{3}{2}$$

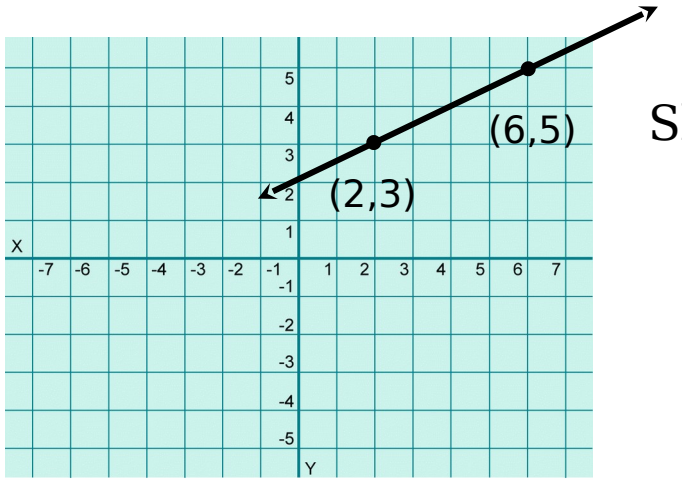
or

$$= \frac{(3-9)}{(5-9)} = \frac{-6}{-4} = \frac{3}{2}$$

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You Try It!

Find the slope between (2, 3) and (6, 5)



$$\text{Slope} = \frac{\text{Vertical Change}}{\text{Horizontal Change}} = \frac{(y_2 - y_1)}{(x_2 - x_1)}$$

$$= \frac{(5-3)}{(6-2)} = \frac{2}{4} = \frac{1}{2}$$

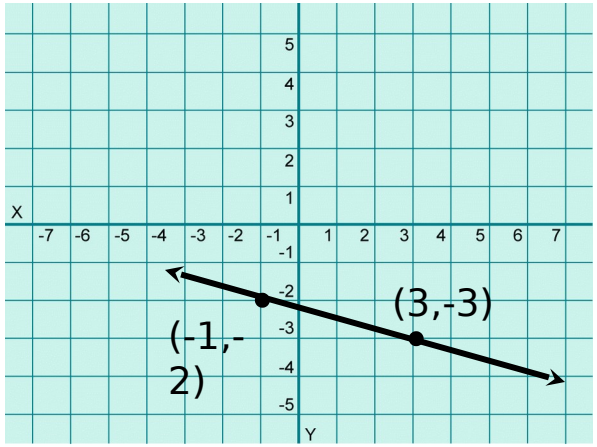
or

$$= \frac{(3-5)}{(2-6)} = \frac{-2}{-4} = \frac{1}{2}$$

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You Try It!

Find the slope between (-1, -2) and (3,-3)



$$\text{Slope} = \frac{\text{Vertical Change}}{\text{Horizontal Change}} = \frac{(y_2 - y_1)}{(x_2 - x_1)}$$

$$= \frac{(-3-(-2))}{(3-(-1))} = \frac{-1}{4} = -\frac{1}{4}$$

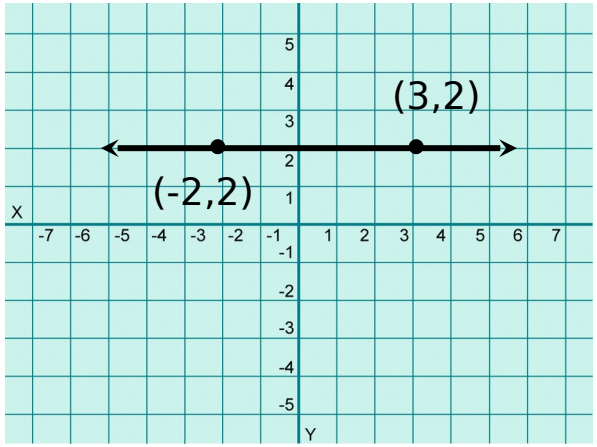
or

$$= \frac{(-2-(-3))}{(-1-3)} = \frac{1}{-4} = -\frac{1}{4}$$

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You Try It!

Find the slope between (-2, 2) and (3,2)



$$\text{Slope} = \frac{\textbf{Vertical Change}}{\textbf{Horizontal Change}} = \frac{(y_2 - y_1)}{(x_2 - x_1)}$$

$$= \frac{(2-2)}{(3-(-2))} = \frac{0}{5} = 0$$

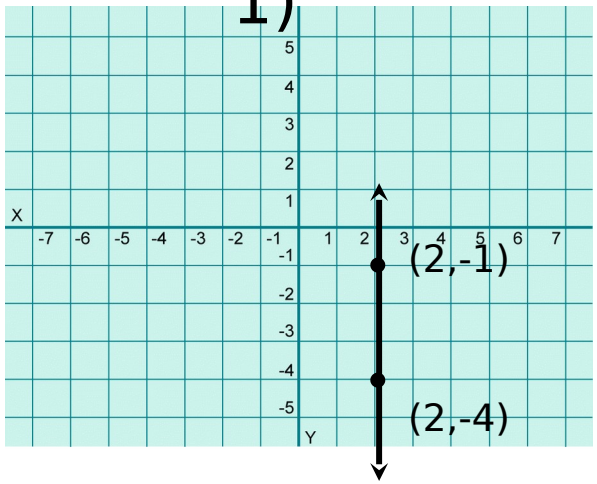
or

$$= \frac{(2-2)}{(-2-3)} = \frac{0}{-5} = 0$$

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You Try It!

1. Find the slope between (2, -4) and (2,-1)



$$\text{Slope} = \frac{\text{Vertical Change}}{\text{Horizontal Change}} = \frac{(y_2 - y_1)}{(x_2 - x_1)}$$

$$= \frac{(-1-(-4))}{(2-2)} = \frac{3}{0} = \textit{undefined}$$

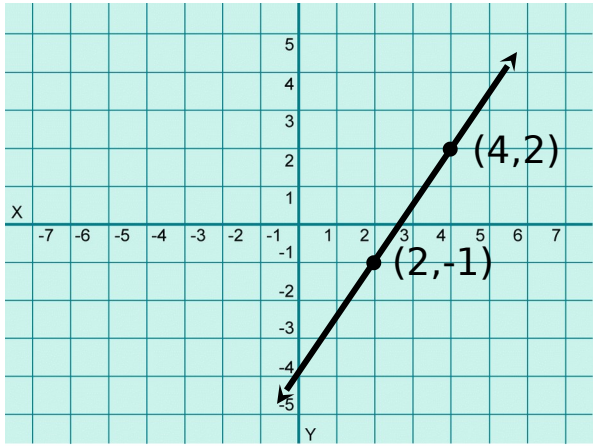
or

$$= \frac{(-4-(-1))}{(2-2)} = \frac{-3}{0} = \textit{undefined}$$

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You Try It!

Find the slope between (2, -1) and (4, 2)



$$\text{Slope} = \frac{\text{Vertical Change}}{\text{Horizontal Change}} = \frac{(y_2 - y_1)}{(x_2 - x_1)}$$

$$= \frac{(2-(-1))}{(4-2)} = \frac{3}{2}$$

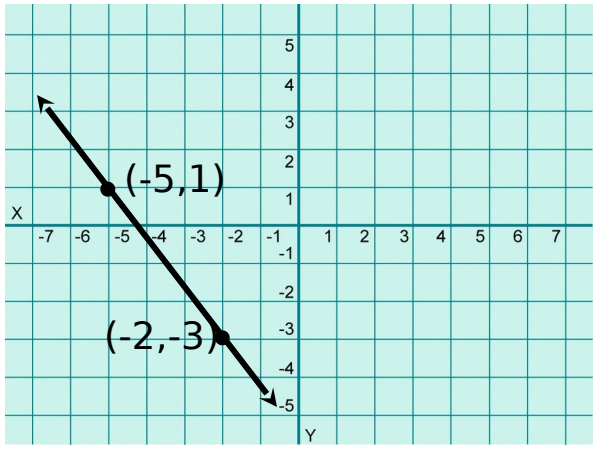
or

$$= \frac{(-1-2)}{(2-4)} = \frac{-3}{-2} = \frac{3}{2}$$

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You Try It!

Find the slope between (-5, 1) and (-2, -3)



$$\text{Slope} = \frac{\text{Vertical Change}}{\text{Horizontal Change}} = \frac{(y_2 - y_1)}{(x_2 - x_1)}$$

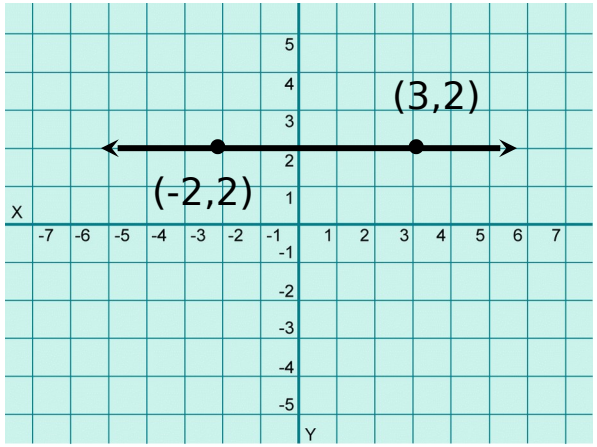
$$= \frac{(-3-1)}{(-2-(-5))} = \frac{-4}{3} = -\frac{4}{3}$$

or

$$= \frac{(1-(-3))}{(-5-(-2))} = \frac{4}{-3} = -\frac{4}{3}$$

You Try It!

Find the slope between (-2, 2) and (3,2)



$$\text{Slope} = \frac{\textbf{Vertical Change}}{\textbf{Horizontal Change}} = \frac{(y_2 - y_1)}{(x_2 - x_1)}$$

$$= \frac{(2-2)}{(3-(-2))} = \frac{0}{5} = 0$$

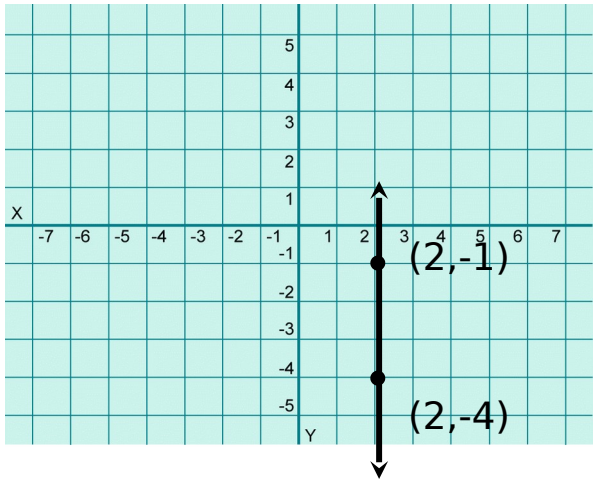
or

$$= \frac{(2-2)}{(-2-3)} = \frac{0}{-5} = 0$$

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You Try It!

Find the slope between (2, -4) and (2,-1)



$$\text{Slope} = \frac{\text{Vertical Change}}{\text{Horizontal Change}} = \frac{(y_2 - y_1)}{(x_2 - x_1)}$$

$$= \frac{(-1-(-4))}{(2-2)} = \frac{3}{0} = \textit{undefined}$$

or

$$= \frac{(-4-(-1))}{(-2-2)} = \frac{-3}{0} = \textit{undefined}$$

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Let’s play a GAME to practice!

Directions

Calculate the slope between the two given points, find it and circle it on your paper faster than your opponent!

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Find the slope between the two points.

$$m = \frac{(y_2 - y_1)}{(x_2 - x_1)}$$

1. (2,4) and (5, 9)
2. (4, 7) and (6, 8)
3. (6, 3) and (7, 5)
4. (3, 0) and (11, 15)
5. (6, 10) and (15, 15)
6. (1, 5) and (5, 8)
7. (2, 1) and (1, 3)
8. (6, 7) and (9, 3)
9. (1, 3) and (4, -7)
10. (9, 3) and (6, 3)
11. (4, 8) and (4, 16)
12. (1, -3) and (4, -2)

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Directions

Calculate the slope between the two given points, find it and circle it faster than your opponent!



1/2

5/8

undefined

-2

15/8

3

2

0

-11/10

5/3

-4/3

-4

17/5

-10/3

1/3

-8/11

4/11

1

5/9

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